KING ABDULAZIZ UNIVERSITY
FACULTY OF COMPUTING & IT

DIYAR

SMART RENTAL MANAGEMENT SYSTEM

DEPARTMENT OF COMPUTER SCIENCE

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GROUP 1

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Introduction

WHY DIYAR?

Traditional rental management is broken and inefficient, leading to high tenant dissatisfaction.

Key challenges include:

- **Poor Communication:** Reliance on disorganized methods like calls and emails.
- **Delayed Maintenance:** Landlords struggle to track requests, causing long delays.
- **Lack of Transparency:** No clear system for payments, records, or tracking progress
- **Unclear payment tracking.**

WHAT IS DIYAR?

Diyar (ديار) is a Smart Rental Management System designed as an integrated digital platform. It makes rental operations seamless, transparent, and accessible.

Project Goal

The goal of Diyar is to modernize and simplify the rental experience by providing a smart, centralized platform that improves communication, automates maintenance tracking, handles emergencies instantly, and enhances transparency through features like real-time chat, AI property recommendations, and rating systems. Ensuring a faster, easier, and more trustworthy rental process.

Objectives

Our system, DIYAR, is designed to achieve several targets:

- Automate communication and maintenance workflows.
- Enhance transparency with real-time tracking and notifications.
- Offer secure online rent payment and reminders.
- Improve decision-making using dashboards and analytics.
- Increase satisfaction for both landlords and tenants.

Problem Statement/Description

Our system, DIYAR , was designed to solve several main problems in the rental process

- Ineffective communication between tenants and landlords that causes delays and misunderstandings
- lack of transparency in maintenance requests, leaving tenants unaware of progress completion
- no unified emergency system to ensure quick response and coordination, like water leaks or power failures.
- Absence of reminders for rent payments and contract expirations, which causes confusion and missed deadlines.
- Rental activities are managed across many separate, uncoordinated platforms instead of unified system

Included Features

- **Property Management:** Allows landlords to add properties with photos, descriptions, and addresses.
- **Communication & Requests:** Direct chat between landlords and tenants for messages and maintenance.
- **Maintenance Scheduling:** Organize maintenance tasks with real-time notifications.

- **Emergency Mode:** One-click emergency alerts for urgent cases.
- **Community Notifications:** Send announcements or reminders to all tenants.
- **Alerts & Reminders:** Automatic rent and contract renewal reminders.

Included Features

- **Ratings & Reviews:** Mutual rating system for both landlords and tenants

- **Interactive Dashboard:**
  1. Landlord Dashboard
  2. Tenant Dashboard:

- **Filter-Based Search:** Search properties by location, price, rooms, and services.

- **Property History:** Shows full maintenance history for transparency.

Excluded Features

- **Legal Procedures:** Excludes legal actions like eviction or dispute resolution.

- **Smart Home Integration:** No smart device control (e.g., lighting, AC).

- **Offline Functionality:** Requires internet connection to work fully.

- **MARKET FORECASTS:** NO PREDICTION OF RENT PRICES OR MARKET TRENDS.

Functional Requirements

User Account:
The system shall allow users to register as tenants or landlords using their full name, email, ID, phone number, Password, and verify via SMS

Login & Security:
The system shall allow registered users to login with ID/password, SMS verification, and password reset via email.

Profile Management:
The system shall allow users to update personal information, upload profile picture, or delete account.

Ratings & Reviews:
The system shall allow both sides to rate and review after a rental period.

Functional Requirements

FOR TENANTS

Property Search:
The system shall allow tenants to search by location, price, type, and amenities, view top-rated properties and get the history maintenance.

Online Payments:
The system shall allow tenants to pay rent online; receive reminders before due date.

Favorites & Contact:
The system shall allow tenants to save preferred properties and contact landlords directly or via system messaging.

Maintenance Requests:
The system shall allow tenants to report issues with images/videos and send emergency alerts.

FOR LANDLORDS:

Property Management :
The system shall allow landlords to add, edit, or delete property listings with full details.

Recieve Maintenance Requests:
The system shall notify landlords of the maintenance requests and emergency alerts.

Nonfunctional Requirements

Performance &Reliability

The system shall Works smoothly on all browsers/devices.

The system shall be available 24/7, and Handles up to 100,000 users concurrently

User Experience & Design

The system shall be Simple, user-friendly interface with light/dark modes and Supports both Arabic & English.

The system shall allow user to complete the Key actions within 3 clicks.

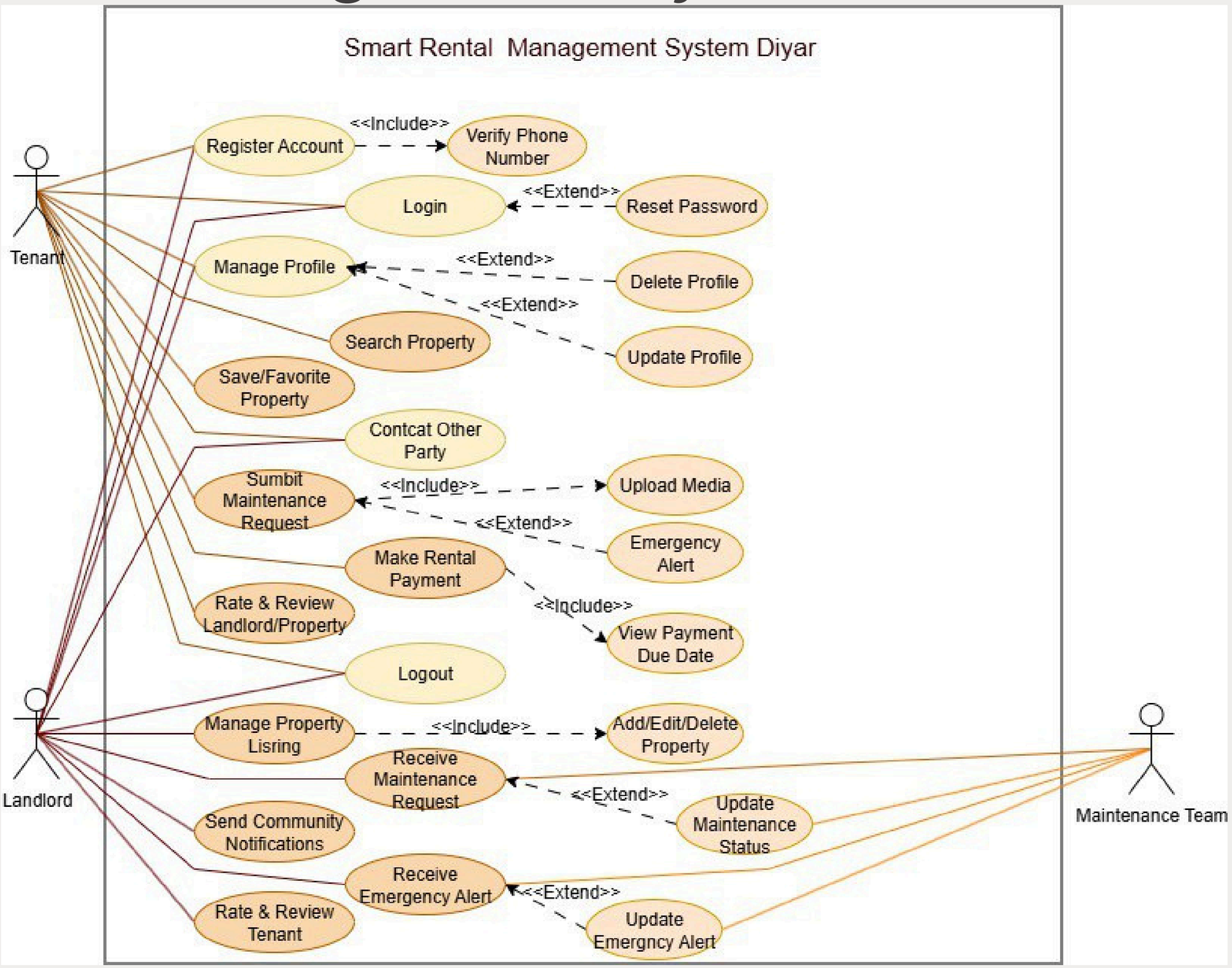
Security & Data

The system shall have strong data encryption and auto-save.

Modular Design

The system shall be easy to maintain and update with a modular design.

Use Case Diagram for Diyar



Use Case Description

FIRST USE CASE DESCRIPTION

USE CASE	SEARCH AND FILTER PROPERTIES (UC-1)
Description	Allows tenants to search for rental properties by applying filters such as location, price, number of rooms, and property type to find suitable options.
Primary Actor	Tenant
Supporting Actors	None
Preconditions	• Tenant must be registered and logged in. • There must be available properties in the system.
Postconditions	• The system displays matching properties. • Tenant can view details, save favorites, or contact the landlord.
Main	• Tenant opens the Search page. • System displays available filters. • Tenant selects preferred criteria. • Clicks “Apply Filters.” • System shows matching results with key info. • Tenant views details, saves, or contacts landlord.
Extensions	• No Filters Selected: System shows all properties. • No Matches Found: System displays “No properties found.” • Save Property: Adds to Favorites list. • Contact Landlord: Opens direct chat page.

Use Case Description

SECOND USE CASE DESCRIPTION

USE CASE	SUBMIT MAINTENANCE REQUEST (UC-2)
Description	Allow the Tenant to submit a detailed maintenance or emergency request to the Landlord and track its resolution.
Primary Actor	Tenant
Supporting Actors	Landlord, Maintenance Service Provider
Preconditions	• The Tenant must be registered and logged into the system. • Tenant must have an active lease linked with a property in the system. • The system must be connected to the Maintenance Team.
Postconditions	• The maintenance request is logged into the system. • The Landlord is notified of the request. • The Tenant is able to view the request status in their dashboard.
Main	• Tenant goes to request section and selects "Submit New Request." • Tenant details the issue, sets Priority, and uploads Attachments. • Tenant clicks "Submit Request." The system validates fields, records the status as "Pending," and notifies the Landlord. • The Landlord changes the status, and the Tenant's view is updated immediately.

Use Case Description

SECOND USE CASE DESCRIPTION

USE CASE	SUBMIT MAINTENANCE REQUEST (UC-2)
Extensions	• **Emergency Mode:** Tenant clicks a one-click alert, and the system instantly notifies the Landlord and Maintenance Service Provider simultaneously. • **Invalid Input:** System displays an error if required fields are missing, returning the Tenant to step 3. • **Landlord Action:** Landlord reviews and updates status (Pending → In Progress → Completed).

Domain Sources Analysis

SIMILAR EXISTING SYSTEMS

Studied platforms like **Booking**, **Gathern**, and **Airbnb** to identify gaps in communication, maintenance tracking, and transparency.

END USERS

Collected **feedback** from landlords and tenants via a **survey** to highlight issues such as slow maintenance and poor communication.

Thank You for Listening

Diyar Smart Rental System

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